

On January 30, 2026, SpaceX
applied for permission to launch
and operate up to one million satellites.

Years of work. One mission.

Out here, every signal matters.



Small failures. Expensive consequences.

59%

Attitude / orbit control
and power systems

156 on-orbit failures · 1980–2005

Tafazoli · Fig. 1

\$4.4B

Reported losses from
power-related failures

64 incidents · 1990–2006

Landis et al. · 2006

Satellite telemetry changes constantly.
Which windows need closer inspection?

- Telemetry: What changed?
- Command context: Was a command executed nearby?
- Our task: Flag potentially anomalous windows for review.

Data used for training

ESA Mission1 · channels 41–46 · 6 h · TimeNet → OpenTSLM

Telemetry

————→ Normalized signal → encoder

What changed?

Telecommands

————→ Command timing → text context

Was a command executed nearby?

ESA annotations

————→ Training targets + evaluation

What was the event labelled as?

OpenTSLM improves on the two-shot LLM

224 test windows · event-grouped · reported accuracy

50.45%

Two-shot Llama-3.2-3B

Recall 0.9%

75.89%

OpenTSLM · retrained

Recall 64.3% · F1 0.727

Classical summary statistics still lead: 90.18% accuracy.

OpenTSLM: 72 positive windows detected · 40 missed · 14 false alarms.

Classical, prompt features: F1 0.891 · OpenTSLM precision: 0.837.

Mission Control

